

CMPDI DETAILED PROJECT REPORT (DPR) - Samaleswari Opencast 15MTY

Subsidiary: MCL | Capacity: 15.0 MT | Geological Basin: Ib Valley Coalfield, Odisha

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This Detailed Project Report (DPR) formulates the mine planning, equipment sizing, and geological stripping layout for Samaleswari Opencast 15MTY under MCL.

Geological Reserve & Setting: Situated in the Ib Valley Coalfield, Odisha, the mine exploits thick coal seams belonging to the Barakar Formation. Total geological reserves are estimated at 480 Million Tonnes.

Rated Production Capacity: 15.0 MT under mechanized Opencast operations. Stripping ratio is projected at 1.25 m3/t, utilizing 42 m3 electric rope shovels, 240T rear dump trucks, and high-capacity bucket wheel systems.

Environmental Mitigation & FMC: Coal evacuation is planned via a First Mile Connectivity (FMC) rapid loading silo with in-pit crushing and a 4.2 km overland conveyor to eliminate surface haul truck emissions.

Technical Parameter	Project Specification	DGMS / Ministry Compliance
Subsidiary / Mine Name	MCL - Samaleswari_Opencast	Approved by CIL Board
Rated Annual Capacity	15.0 MT	Environmental Clearance Granted
Mining Method	Opencast	High-wall Stability Factor > 1.3
Stripping Ratio	1.25 m3/t	Continuous Reclamation Mandated
Evacuation Mode	Rapid Loading Silo / FMC	Zero Surface Dust Spillage